

Emel YILDIRIM

Ph. D. Physics/Physics teacher

I am a physics teacher with a PhD in physics. I work at a public school for gifted students. My doctoral research focused on semiconductor thin films and nanomaterials. I have a keen interest in astronomy. I work on projects across various fields with students of all ages. Recently, I have been interested in data science and artificial intelligence, and I am taking courses in these fields.

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SKILLS

Physics Teacher
Semiconductor

Thin Films
Science Projects

Education
Data Science

STEM/STEAM
Astronomy

Programming
lang-AI

EDUCATION

Ph. D. Physics

Mersin University 2023

Thesis: "OBTAINING AND CHARACTERIZATION OF p TYPE Si AND CONDUCTIVE SUBSTRATED CdS HETEROJUNCTIONS BY CHEMICAL BATH DEPOSITION METHOD"

Master's Degree in Physics

Mersin University 2013

Thesis: "CHARACTERIZATION of ZnSe SEMICONDUCTOR THIN FILMS "

Undergraduate Education

Hacettepe University 2001

Physics Teaching (German)

EXPERIENCE

Book Publications

- E. YILDIRIM & M. A. YÜKSEL, Aşık Veysel'in Gözünden ve Gönülünden Atatürk ve Cumhuriyet, H. AKDAĞ, İ. Y. ÖZTÜRK & F. S. SAY [Editörler], Eğitimde ve Kültürde Akademik Araştırmalar (222 - 236), ISBN: E-ISBN:978-625-5980-090-0), TÜRKİYE: Kitap Dünyası, 01 Aralık 2025, Kitapta Bölüm.
- E. ŞAHİN, M. YÜKSEKSÖYLEMEZ, D. POLAT, E. YILDIRIM, H. KÜÇÜKLER & N. TUNALI, Bilim ve Sanat Merkezleri Fizik Alanı Yardımcı Ders Materyali, ISBN: 978 975-11-5830-7: Sude Ajans Reklam Org. Tan. LTD. ŞTİ, 01 Mart 2025, Kitap.

Journal Publications

- Yildirim, E., Ozmen, S. I., Havare, A. K., & Gubur, H. M. (2026). Substrate effect on nanowall-structured CdS thin films obtained by CBD: Structural, optical, and electrochemical characterization for next-generation optoelectronic applications. *Materials Chemistry and Physics*, 348, 131651.
- Yildirim, E., Ozmen, S. I., Havare, A. K., and Metin Gubur, H. (2023). Annealing Effect on CdS Nanowalls Grown by Chemical Bath Deposition on Glass Substrate. *Physica Scripta*. Vol. 98, 075933.
- Yildirim, E.; Metin Gubur, H.; Alpdogan, S.; Ari, M.; Harputlu, E.; Ocakoglu, K. The effect of annealing of ZnSe nanocrystal thin films in air atmosphere. *INDIAN JOURNAL OF PHYSICS*, 2016, 90, 793-803.

Oral Presentations

- Baran, E., Can, O.O., Ateş A. K., Yıldırım, E., "Muz Lifinden Isı Yalıtım Malzemesi" International Science and Art Student Congress (ISASC) 2026, 8-10 Jun, 2026, Muğla, Türkiye.
- Akyürek B., Okuyan G., Yıldırım E., "Bölge Bölge Türkülerin Yâre Verilen Kıymet Üzerinden Aile Yapısını Koruyucu Değerlere Ait İçerik Analizi" International Science and Art Student Congress (ISASC) 2026, 8-10 Jun, 2026, Muğla, Türkiye.
- Yıldırım, E., Yüksel M. A.; "Atatürk And The Republic From The Eyes And Heart of Aşık Veysel" 4th International Symposium On Academic Studies In Education And Culture (ISASEC 2024), 19-21 Eylül 2024 Anamur, Mersin, Türkiye.
- Yıldırım E., İldan Özmen S. , Metin Gübür H., Havare A. K., Characterization of CdS Nanowalls Deposited on Glass, ITO, FTO and p-Si Substrates by Chemical Bath Deposition, 4th International Natural Science, Engineering and Material Technologies Conference-NEM 2024, September 12-14 2024, Kırklareli, Türkiye.
- Yıldırım, E., Bilgin, H. S., "STEAM Approach to the Poisson Point" 5th International STEM Education Conference, July 6-7, 2024, İstanbul İbn Haldun University, İstanbul, Türkiye.
- Yıldırım, E., Bağırhanlı H. U., Kütük M. K., "Effect of Electric Field on The Development of Onion Plant", 5th International STEM Education Conference, July 6-7, 2024, İstanbul İbn Haldun University, İstanbul, Türkiye.

7. Yavuz Yolcu E., Atlen M., Yıldırım E., Öniş D., "Creating Multiple Picture Paintings with Lights Using Pictures as Color Filter" 5th International STEM Education Conference, July 6-7, 2024, İstanbul İbn Haldun University, İstanbul, Türkiye.
8. Yıldırım, E., Şahin Y.E., "Türk Astronot Alper Gezeravcı'nın Uzaya Gitmesinin Silifke Halkında Uyandırdığı Duygular/Değerler", VII. Ulusal Çocuk Araştırmaları Kongresi, 12-14 Haziran 2024, Yozgat, Türkiye.
9. Teker E., Yıldırım E., "Enis Tataroğlu Who Knows 58 Languages: Language Learning Stories and Ways", 2st International Positive Schools And Well-Being Congress "Lifelong Learning and Well-Being in a Digitalized World", 29-31 May 2024, Girne, K.K.T.C.
10. Büyükkaya M. C., Yıldırım, E., "Cumhuriyet'in Son Tanıklarından Sümerolog Muazzez İlmiye Çığ'ın Gözünden Atatürk, Cumhuriyet ve Türk Gençliği", Bilim Armonisi Uluslararası Gençlik Kongresi" 15-16 Aralık 2023, Antalya, Türkiye.
11. Güneş E. F., Yıldırım, E. "Değerler Eğitiminde Sinemanın Etkisi - Cep Herkülü: Naim Süleymanoğlu Örneği Bilim Armonisi", Bilim Armonisi Uluslararası Gençlik Kongresi" 15-16 Aralık 2023, Antalya, Türkiye.
12. Mengilli, E., Yıldırım, E. Muazzez İlmiye Çığ'ın Bir Asır Sağlıklı Yaşam Sırlarının Değerler Eğitimi Bağlamında Değerlendirilmesi Bilim Armonisi Uluslararası Gençlik Kongresi" 15-16 Aralık 2023, Antalya, Türkiye.
13. Yıldırım, E.; Özmen S. İ.; Havare A. K.; Gübür H. M. "Growth and Characterization of CdS Nanoflowers by Chemical Bath Deposition and Effect of Annealing" Sözlü bildiri, "9th International Conference on Materials Science and Nanotechnology for Next Generation (MSNG-2022)", September 22-24, 2022, Gazi University, Ankara, Türkiye.
14. Yıldırım, E., Gübür, H. M. "Characterization of polycrystalline p-si/CdS thin films prepared by chemical bath deposition" Sözlü bildiri, "9th International Conference on Materials Science and Nanotechnology for Next Generation (MSNG-2022)", September 22-24, 2022, Gazi University, Ankara, Türkiye.
15. Yıldırım, E., Çelik, A. "Farklı Kağıt Uçak Modellerinin Uçuş Mesafesi, Uçuş Süresi ve Hızlarının İncelenmesi" Sözlü bildiri, VI. Ulusal Çocuk Kongresi (UÇAK), 08-10 Eylül, 2022, Konya, Türkiye.
16. Yıldırım, E.; Metin Gübür, H. "Production of CdS thin films by chemical bath deposition method". Sözlü bildiri, 4th International Conference on Physical Chemistry and Functional Materials (PCFM21), 08-09 April 2021 Firat University, Elazığ, Türkiye.
17. Yıldırım, E.; Ünlü, B.; Yılmaz, M., Measurement of radon gas in banana greenhouses in the province of Mersin, district Bozyazı. TFD 33. Uluslararası Fizik Kongresi, 2017-09-06, 2014-09-10, Bodrum-Muğla, Türkiye, 2017.

Poster Presentations

1. Çıtır, M., **Yıldırım, E.**, Türkiye. Sümerolog Muazzez İlmiye Çığ: "Sümerliler Türklerin Bir Koludur" Bilim Armonisi Uluslararası Gençlik Kongresi" 15-16 Aralık 2023, Antalya.
2. **Yıldırım, E.**, Gübür H. M. "Characterization of Glass/FTO/CdS Thin Films by Chemical Bath Deposition" Poster bildiri, "Turkish Physical Society-38th International Physics Congress (TPS-38)", August 31-September 4, 2022, Bodrum, Türkiye.
3. **Yıldırım, E.**, Gübür H. M. "Structural, Optical and Electrical Properties of CdS Thin Films on Glass/ITO" Poster bildiri, Turkish Physical Society-38th International Physics Congress (TPS-38), August 31-September 4, 2022, Bodrum, Türkiye.
4. **Yıldırım, E.**, Kurt, F., Kart, Y. "Radon measurement in the cafeteria and chemical laboratory of Bozyazı Highschool." Poster bildiri. TFD 33. Uluslararası Fizik Kongresi, 2017-09-06, 2017-09-10, Bodrum-Muğla, Türkiye.
5. **Yıldırım, E.**, Metin Gübür, H., Alpdoğan, S. "The effect of annealing temperature on the structural, optical and electrical properties of ZnSe Films." Poster bildiri, TFD 31. Uluslararası Fizik Kongresi, 2014-07-21, 2014-07-24, Bodrum-Muğla, Türkiye.

SOME OF THE OTHER ACTIVITIES AND PROJECTS

Intro2Astro2026

Scientific research project supported by higher education institutions, Researcher, "Preparation and Characterization of CdS Thin Films on p-type Si and Conductive Substrates via Chemical Deposition Method," Executing Institution: MERSIN UNIVERSITY, Supporting Institution: MERSIN UNIVERSITY (Domestic), June 1, 2020 – October 1, 2023.

BAP-FBE FB (EY) 2012-2 YL "Characterization of ZnSe Semiconductor Thin Films," Executing Institution: MERSIN UNIVERSITY, January 1, 2012 – July 1, 2013.

CERN 2nd Turkish Teacher Programme, July 28–August 1, 2014, Geneva, Switzerland.

Nobel Prize Teacher Summit, March 2025, Stockholm, Sweden.

HERTZ: Summit on Communication, Education, and Radio Technologies for BİLSEM Teachers (Participant), TÜBİTAK 4005, September 1–6, 2025, Samsun.

The Mystery of the Nucleus: Nuclear Energy and Radiation Awareness at AIOLIS, October 7–11, 2024, Manisa.

Workshop on Revising Science and Art Center Programs and Preparing Supplementary Course Materials, July 12–16, 2021.

MASDIV (Supporting Mathematics and Science Teachers) Project, Erasmus – Hacettepe University Mersin Provincial Directorate of National Education, February 2, 2020.

Workshop on Educator Training for the Gifted, Ministry of National Education (MEB), December 11–15, 2017.

Mathematics and Science for Life (MASCIL) Teacher Training Workshop, Hacettepe University, June 29, 2016. TÜBİTAK Course on Scientific Project Mentorship for Teachers (Adana Regional Center Provinces) (June 11–14, 2012 – 32 hours) / Mersin University.

TÜBİTAK Project Mentorship Training Workshop for High School Physics Teachers (February 2–8, 2014) / Süleyman Demirel University.

Future English (Erasmus Project), Ministry of National Education (MEB), April 2025, participant, Italy.

Bartın Science School: Explore, Learn, and Have Fun in Nature; Instructor; 4004 - Nature Education and Science Schools Program, 2024.

Participation with 5 students, Turkish Space Travelers Week, January 29 – February 3, 2023, TUA (Turkish Space Agency), MEB, Space Camp, İzmir.
International Science Night (Mersin), 2019, Workshop Leader.
Mersin Science Festival (TÜBİTAK 4007), 2019, Workshop Leader.
International Science Night (Mersin), 2019, Advisor.
Mersin Science Festival (TÜBİTAK 4007), 2019, Advisor.
2017-TÜBİTAK 4007, "Traveling Science on the Road" (Gezen Bilim Yolları), Guide.
Robotics Goes to Europe (Erasmus+ Project), MEB, 2018, Participant, Turkey.
Physics, Science, and the Universe (2018), school-based project, coordinator.
International Save Our Species (SOS) Project (2018–2019), school coordinator.
Nature and Science Academy (2019), school-based project, coordinator.
Science and Art in Nature (2019), school-based project, workshop leader.
The Astronomy Journey Begins with BİLSEM (2019), school-based project, workshop leader.
TÜBA 7th Applied Science Education Course, Mersin University, January 27–31, 2020, mentor.
STEM Makers Fest/Expo, Hacettepe University (Gültekin Çakmakçı) / Nesibe Aydın Mersin Schools, February 8, 2020, workshop leader.

AWARDS

Mersin Provincial Representative for Teachers' Day (Formerly known as Teacher of the Year)

Scratch Cup 2026

The advisor of the team that placed first (Best Screenplay Award)

Certificates of outstanding achievement and letters of appreciation in teaching

LANGUAGES

Turkish

Mother tongue

English

Intermediate

German

Basic-intermediate